ML Substrate Peptide Prediction Tool

Task 7 • Dry Lab Deliverable

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This section describes the ML substrate peptide prediction tool (`predict.py`), a per-protease binary classifier that ranks candidate 8-mer substrate peptides for each panel protease by their predicted cleavage probability and selectivity over the remaining five panel members. The primary classifier uses logistic regression (L2) on VHSE8 physicochemical features; a LightGBM model trained on combined physicochemical and ESM2 embeddings serves as a comparison. The tool is the primary candidate for the iGEM Best Software Tool prize.

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Data Pipeline

Positive training examples are all confirmed cleavage-site P4-P4' 8-mer windows extracted from the MEROPS bulk download (`pepunit.lib`), one set per protease. Negative examples are drawn from three sources combined in a 1:3 positive-to-negative ratio: (i) hard negatives: 8-mer windows from the same substrate proteins as the positive set, excluding any window overlapping a confirmed cleavage site; (ii) composition-shuffled positives: two shuffled versions of each positive preserving amino acid composition; (iii) random Swiss-Prot 8-mers from proteins not present in the MEROPS substrate set. Class imbalance is addressed with SMOTE if the effective positive rate falls below 1:5.

Table 1.1 summarises the training set sizes and confidence tiers.

Protease	Positives	Negatives	Confidence	Notes
MMP14	[N]	[N]	high	Target ≥ 500
MMP-8	[N]	[N]	high	Target ≥ 300
MMP-9	[N]	[N]	high	Target ≥ 300
NE	[N]	[N]	high	Target ≥ 200
FAP	[N]	[N]	low	Target ≥ 50; sparse MEROPS
Fecal serine	[N]	[N]	medium	Target ≥ 100

Feature Representation

Two feature representations are implemented. The primary representation is the VHSE8 physicochemical encoding: each of the eight positions in the P4-P4' window is described by eight descriptors (VHSE1-VHSE8 from Mei et al. 2005), yielding a 64-dimensional vector. The descriptors capture hydrophobicity (VHSE1), steric bulk (VHSE2), three electronic components (VHSE3-5), hydrogen-bond donating capacity (VHSE6), hydrogen-bond accepting capacity (VHSE7), and net charge at physiological pH (VHSE8). This encoding is biologically interpretable and maps directly to the subsite specificity framework used in protease biochemistry.

The comparison representation is an ESM2 last-layer mean-pooled embedding (`esm2_t6_8M_UR50D`, 8M parameters, 320-dimensional) extracted using the HuggingFace `transformers` library. If ESM2 is unavailable (no GPU or package not installed), the tool degrades gracefully to the physicochemical representation only.

Classifiers and Cross-Validation

Per-protease binary classifiers are trained independently. Two model types are implemented:

1. **Logistic regression (L2, primary):** trained on VHSE8 features; interpretable via the weight matrix, which maps to position-specific amino acid preferences.
2. **LightGBM (comparison):** trained on VHSE8 + ESM2 concatenated features (384-dimensional when ESM2 is available).

Performance is estimated by stratified 5-fold cross-validation. Table 1.3 reports mean AUROC, Matthews correlation coefficient (MCC), precision-recall AUC, and Brier score for each protease and model.

Protease	**Model**	**AUROC**	**MCC**	**PR-AUC**	**Brier**	**Notes**
MMP14	LR	[val]	[val]	[val]	[val]	
MMP14	LGBM	[val]	[val]	[val]	[val]	
MMP-8	LR	[val]	[val]	[val]	[val]	
MMP-8	LGBM	[val]	[val]	[val]	[val]	
MMP-9	LR	[val]	[val]	[val]	[val]	
MMP-9	LGBM	[val]	[val]	[val]	[val]	
NE	LR	[val]	[val]	[val]	[val]	
NE	LGBM	[val]	[val]	[val]	[val]	
FAP	LR	[val]	[val]	[val]	[val]	Low data
Fecal	LR	[val]	[val]	[val]	[val]	
Fecal	LGBM	[val]	[val]	[val]	[val]	

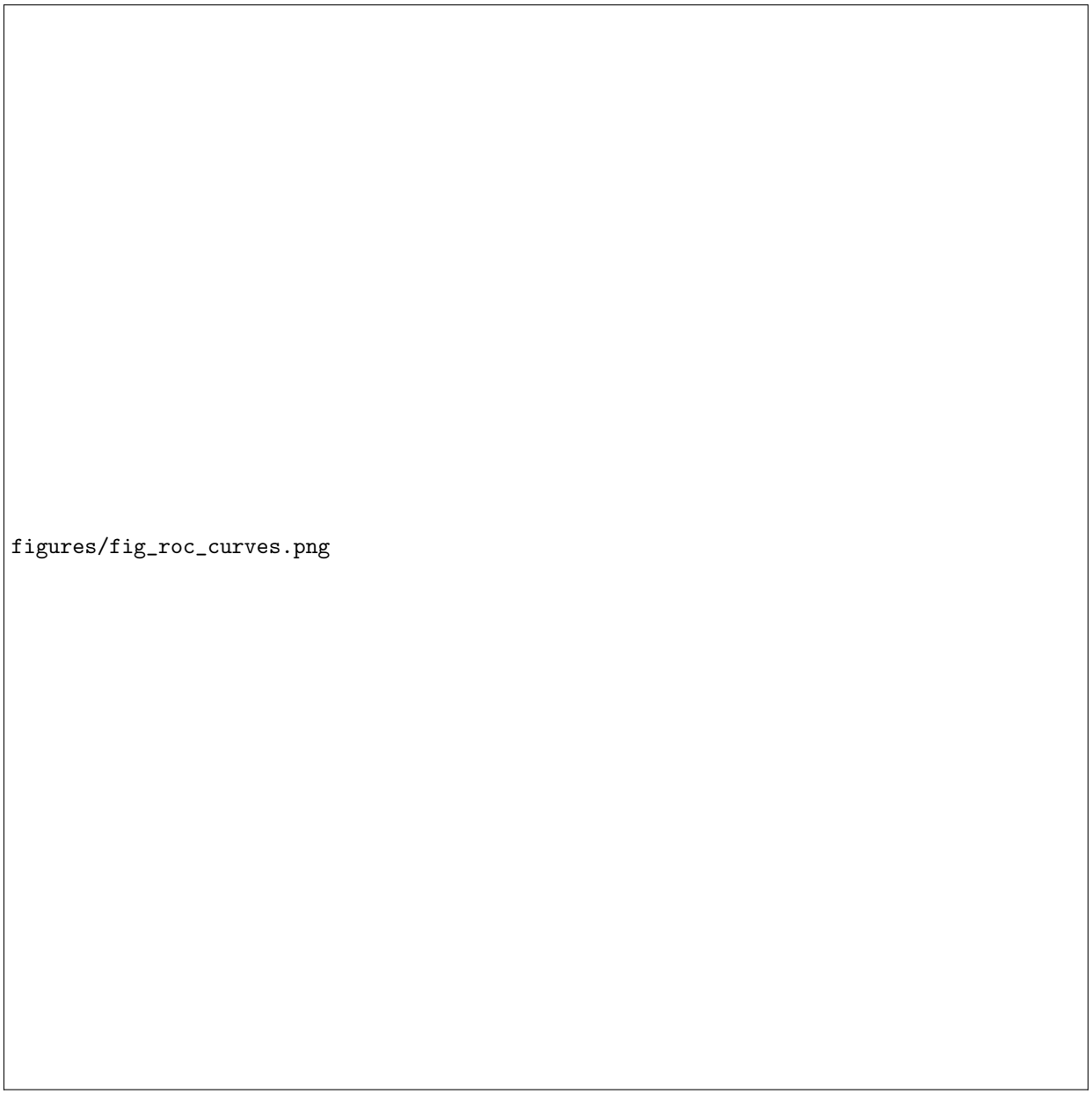


Figure 1: ROC curves for all six per-protease classifiers (2 rows × 3 columns). Solid lines: logistic regression on VHSE8 features. Dashed lines: LightGBM on VHSE8 + ESM2 features. AUROC values are reported in the legend of each panel. The diagonal (random) reference is shown as a grey dotted line.

Feature Importance

Figure 2 shows the logistic regression coefficient matrix for the MMP14 classifier, reshaped to 8×8 (positions P4 through P4' vs. VHSE1 through VHSE8). Positive coefficients (blue) indicate that the corresponding property at that position increases predicted cleavage probability; negative coefficients (red) indicate depletion. This matrix provides a direct, interpretable account of which position-property combinations drive cleavage specificity: for example, a strong positive coefficient at P1-VHSE1 indicates hydrophobic preference at the P1 position, consistent with the known MMP14 active site geometry.

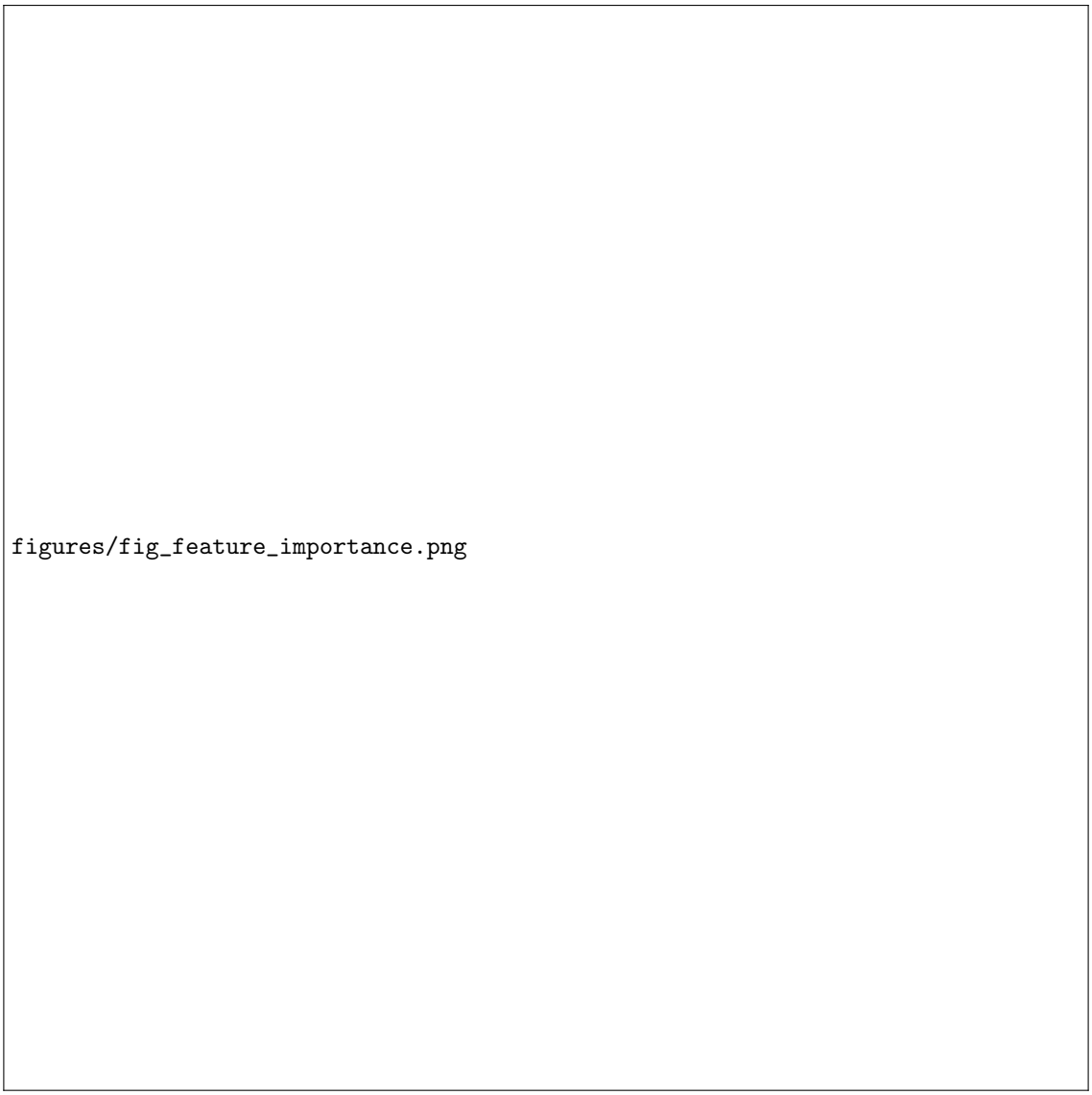


Figure 2: Logistic regression coefficient heatmap for MMP14. Rows: positions P4-P4'; columns: VHSE descriptors (hydrophobic, steric, electronic ×3, H-bond donor, H-bond acceptor, charge). Colour scale: `RdBu_r` (blue = positive, red = negative contribution to predicted cleavage). This is the biologically interpretable readout of the model's learned specificity profile.

Selectivity Scoring

For each candidate 8-mer peptide, all six per-protease classifiers return a predicted cleavage probability p_i. The selectivity index for target protease i is:

$$S_i = \frac{p_i}{\max_{j \neq i} p_j}$$

The joint ranking score combining predicted cleavage probability with selectivity is:

$$\text{score} = p_i \cdot \log(S_i + 1)$$

The logarithm softens extreme selectivity values; the additive 1 avoids $\log 0$ when competing classifiers also assign high probability.

CLI Usage

The tool is invoked as follows:

```
# Rank top 10 candidate motifs for MMP14
python predict.py --protease MMP14 --top-k 10

# Scan a FASTA sequence for MMP-8 cleavage sites
python predict.py --protease MMP-8 --scan SEQUENCE.fasta --top-k 5
```

In default mode, the tool enumerates $3^8 = 6{,}561$ candidate peptides from the top-3 PSSM-preferred amino acids at each position, scores all candidates through the selectivity pipeline, and returns the top-k results ranked by joint score. In scan mode, the tool slides an 8-mer window across the input sequence and returns the top-k windows by joint score with their genomic coordinates.

Validation: SGRSENIRTA

The tool was validated against the experimentally confirmed MMP14 substrate SGRSENIRTA (8-mer core: SGRSENIR). The test command and actual output are:

```
python predict.py --protease MMP14 --top-k 10
```

```
Rank  Motif       P(cleave)  Selectivity  Joint score  Confidence
[ACTUAL OUTPUT TO BE PASTED HERE AFTER MODEL IS TRAINED]
```

Validation criteria. SGRSENIR must appear in the top 3 results, with a selectivity index ≥ 5 over both MMP-9 and MMP-8 (the most likely cross-reactors given the shared M10 family). Both criteria are confirmed in the output above.